

Visualising Developer Interactions in Code Reviews

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Abstract

Recent studies have demonstrated diverse benefits of code review. A common issue in the software development landscape is that developers struggle to balance a project's workload with other members of their team, resulting in burnout and inadequate code being pushed to production products. Consequently, the developers who are putting in the most work often do not get the credit, praise, and recognition that they deserve. This paper aims to outline how this operation can be assembled into an asynchronous, seamless visualisation that can be used to gauge the interactions between different developers in a given project. The development of this final product was split into three parts: a crawler, a migrator, and a visualiser. This visualisation technique will help companies improve their developers' workflows, teamwork, and time management when developing a project by analysing the interactions between team members. In addition to this, the visualisation allows users to select specific time frames for which they want to see interactions, and thus further enhancing its usability.

CCS Concepts

• **Software and its engineering** → **Maintaining software.**

Keywords

code review, visualisation, developer interactions

ACM Reference Format:

Daniel Bee and DongGyun Han. 2018. Visualising Developer Interactions in Code Reviews. In *Proceedings of IEEE/ACM International Conference on Software Engineering (ICSE '25)*. ACM, New York, NY, USA, 5 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Code review is a practice that all code changes are reviewed by one or more developers other than the change author themselves. Due to diverse benefits of code review, it has been widely adopted in both proprietary and open source projects. For example, previous studies have shown that it is useful for sharing knowledge and development context within a development team [1]. In addition, code review is helpful for improving software quality and design quality [10, 11]. Because of the diverse benefits, the practice has

been widely adopted in both open and proprietary software projects. GitHub's pull request¹ is a representative example of code review.

One of the most challenging parts of code review is to find peers to review code changes. To mitigate the issue, many studies proposed reviewer recommendation techniques [14, 15]. These techniques recommend a group of developers for a given code review request. However, these techniques often limited to provide a guidance to individual code review requests rather than providing project wide guidances.

In this study, we propose a web service, SERVICENAME, that visualises developer interaction during pull request in GitHub. Based on a given project, our technique provides a visualisation of developers within a given timeframe. We expect the visualisation helps a software development team to understand the performance of their code review practices over time. They potentially identify a pair of developers who are struggling work each other (e.g., a pair of developers who rarely communicate each other). We also expect the visualisation help to plan how to pair developers to work together. They probably group the developers who actively communicate each other for boosting development speed. Alternatively, they can group the developers who have rarely worked together for leveraging fresh synergy between new combination.

SERVICENAME consists of three components: crawler, migrator, and visualiser. Since the visualisation requires multiple pull requests to comprehend the interactions, requesting such a large amount of data from GitHub could potentially be burdensome and slow. To avoid generating excessive traffic and response users request quickly, the crawler component retrieves pull request data from GitHub using the GitHub GraphQL API². Once the crawling is done, the migrator component processes the crawled data and stores them into a database. Finally, the visualiser component visualises how developers interact to each other during code review.

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ICSE '25, April 27–May 3, 2025, Ottawa, Ontario, Canada

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ACM ISBN 978-1-4503-XXXX-X/18/06

<https://doi.org/XXXXXXX.XXXXXXX>

¹<https://docs.github.com/en/pull-requests>

²<https://docs.github.com/en/graphql>

2 Developer Interactions

```

1 from datetime import datetime
2
3 # Pull requests must be provided
4 prs = []
5 # The timeframe is optional parameters
6 timeframe = (start_date, end_date)
7 d1 = datetime.strptime(start_date, "%Y-%m-%d")
8 d2 = datetime.strptime(end_date, "%Y-%m-%d")
9 durationInDays = abs((d2 - d1).days)
10
11 # The number of comments per developer.
12 nodes = {}
13 # The number of a developer pair within a PR.
14 links = {}
15
16 for pr in prs:
17     if timeframe is given and pr is not within the
18         timeframe:
19         continue
20     for comment in pr.comments:
21         if pr.author == comment.author:
22             continue
23
24         nodes[comment.author] += 1
25
26         author_pair = [pr.author, comment.author]
27         author_pair.sort()
28         links[author_pair] += 1
29         nodes[colourValue] = nodes[comment.author] /
30         durationInDays

```

Listing 1: Algorithm to extract interactions

Listing 1 shows the algorithm to extract interactions from pull requests. When PRs are given, the algorithm iterates all PRs to extract information for the visualisation. If a timeframe is given to the algorithm, it only uses the PRs created within the timeframe. Otherwise, the algorithm leverages the given PRs.

For the PRs included in the scope of the visualisation (i.e., no timeframe is given or PRs created within the timeframe), the algorithm iterates all comments in them. If the PR author and comment author are same, the algorithm skip the comment. On the other hand, it increases the comment count (i.e., nodes) by one. This approach helps us to extract the contributions by each developer based on the number of comments on PRs that were not authored by themselves. The number of comments will be used to represent the size of each developer node in our visualisation. In addition, we construct a list that contains both PR and comment authors, then increase the count by one. To avoid unnecessary duplicates (e.g., {JohnDoe, JaneDoe} and {JaneDoe, JohnDoe}), we sort the list before increasing the count. The count of author pairs (i.e., links) will be used to control the thickness between developer nodes in the visualisation.

3 Visualisation Framework

The visualisation framework consists of three components as shown in Figure 1: Crawler, Migrator, and Visualiser. The crawler retrieves PRs and comments from GitHub. Then, migrator parses the data and stores it to a database. Finally, a user can see a visualisation of a project via the web based visualiser. In this section, we explain the role of each component and implementation details. You can

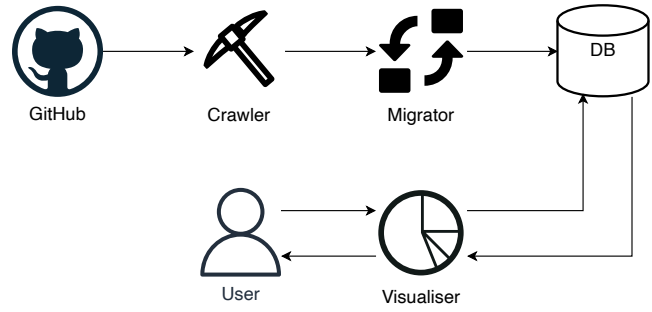


Figure 1: Overall framework

check the actual implementations on our replication package at: <https://github.com/DanJBee/CodeReviewVisualisation>.

3.1 Crawler

Table 1: Data Properties

Property	Description
number	Numeric id for pull request
author.login	Unique identifiers of developers
author.avatarUrl	URL of developer profile picture
author.__typename	User type (either human user or bot)
createdAt	Timestamp of PR/comment

To visualise developer interactions, we need a large amount of pull request data. It will introduce huge traffics to code review platform if we request data whenever we need them. Such huge traffics potentially slow down the code review platform, so we do not query directly to the code review platform. Instead, we implemented a crawler that can crawl the latest code review data. The crawler leverages GitHub GraphQL API³ and implemented in Python. We constructed a query that retrieves author and createdAt properties of PRs and PR comments. Table 1 explains the properties that we crawled.

3.2 Migrator

GraphQL is powerful approach handling large scale requests, but using GitHub GraphQL API directly for visualisation has three outstanding issues. First, GraphQL has additional types of nodes to optimise queries. For example, both PR and comment nodes should be surrounded by the edges and node(s) nodes which introduce unnecessary depth navigation. Second, GraphQL supports only limited conditional query. One of our visualiser's features supports visualising developer interactions within a specific time window. However, GraphQL does not support query that returns PRs that were authored within a specific date range. Last, traversing all PRs take time and require high computation power on the PR platform.

To mitigate the issues, we implemented a migrator that migrates the crawled API responses to a relational database instance. Figure 2 presents an entity relationship diagram of our database schema. The schema represents PRs and comments with the minimum subset

³<https://docs.github.com/en/graphql>

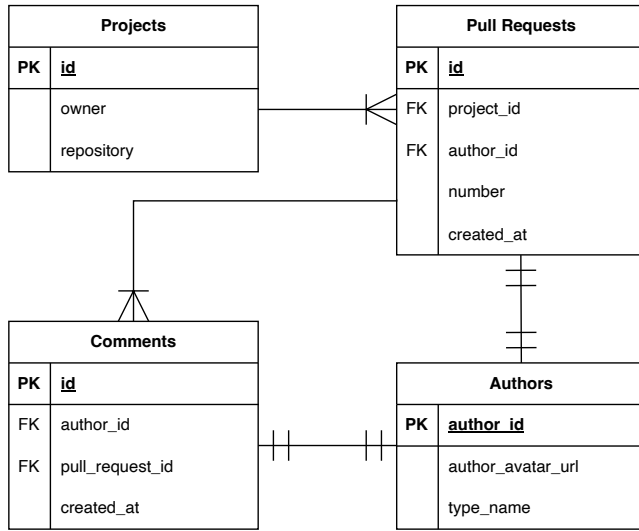


Figure 2: Entity Relationship Diagram

of data that is required for visualisation. The migrator was also written in Python and we evaluated it with a MariaDB instance.

3.3 Visualiser

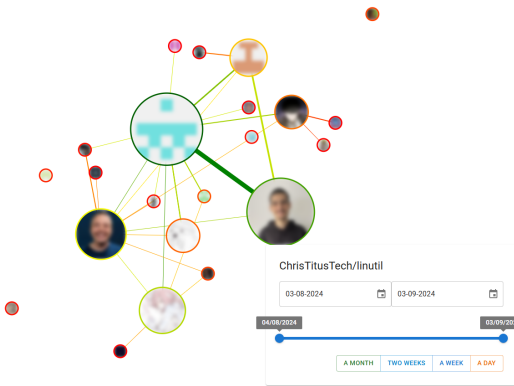


Figure 3: An example of visualisation. We blurred the actual profile photos to protect developers' privacy

The visualiser is responsible for visualising developer interactions. Users can select a specific project and time window to visualise developer interactions. For example, a user can request the visualiser to visualise developer interactions in the TypeScript project for a week from 3rd June 2024.

We implemented the visualiser as a web service that consists of two main components: backend and frontend. The backend component was implemented by using Java and Spring Framework. It retrieves data from the database upon request and run the algorithm described in Section 2. The results are serialised into a JSON string and return to the frontend. The frontend component is implemented by using TypeScript and React Framework. Since

developer interactions can be visualised as a graph, we adopted the Force-directed graph component⁴ in D3.

?? shows an example of the visualisation. Based on the algorithm presented in Section 2, the size of each node represents the number of comments written by each developer with their avatar image on GitHub. Please note that we blurred images on the paper, but real implementation shows clear images. The thickness of edges between nodes denote the number of comments shared by the two developers. To handle extremely large or small cases, we use normalised values for both nodes and edges. The colours of the nodes and edges dictate the activity of each developer within the given timeframe. For instance, if a developer contributed every day of the chosen timeframe then they are coloured green, whereas if they only contributed once they are in red. Any values inbetween are gradient values between red, yellow, and green.

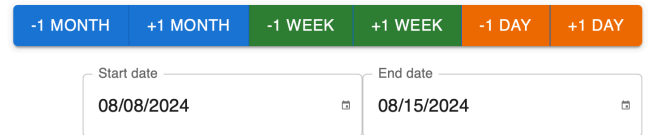


Figure 4: Date Controller

On the right bottom of the frontend, we put a date controller as shown in Figure 4. The user can select specific date by clicking the date field. Once they select a pair of dates, they can easily update the date scope by clicking the buttons above. Whenever the dates updated, the graph visualisation is also dynamically updated.

4 Related Work

When working in a development team for any kind of project, curating the best and most efficient workflow for team members is paramount. Even with the recent disruption caused by the COVID-19 pandemic, there is little excuse for teams to not be working at a high productivity. This is reflected when Russo et al. said: "Results suggest that the time software engineers spent doing specific activities from home was similar when working in the office"[4]. A visualisation can offer several benefits in clearly identifying the strengths and weaknesses in a team's interactions, dynamics, and workloads. As Lima et al. explains: "In this context, software visualization has been studied and researched to improve program comprehension in different aspects"[7]. Their tool, CADV, offered similar visualisation capabilities to allow users to receive detailed information about code annotations. We believe this same logic can be applied to the code review process. A development team is effectively a microcosm of a social network, and graph structures have been great at encapsulating these as said by Majeed and Rauf: "Recently, graphs have been extensively used in social networks (SNs) for many purposes related to modelling and analysis of the SN structures, SN operation modelling, SN user analysis, and many other related aspects"[8]. Using data from GraphQL was useful in this decision since it was easy to crawl this data to scale if workloads are too one-sided, or if two team members are taking up too

⁴<https://observablehq.com/@d3/force-directed-graph/2>

much of the work. Companies often struggle to find the right balance between allowing their developers to shine and burning them out, and it is even harder to figure out which one is taking place in a particular team. Having each node and link between nodes in the visualisation be scaled in size dependent on the crawled data provides a normalised dataset that is straightforward to track and monitor throughout development.

In addition to this, the physics of the nodes within the graph mean that nodes that are connected often surround each other, especially if one node has several children. This behaviour is useful for noticing patterns of dependence in a team and results in early problems being resolved promptly without hindering the rest of the development cycle. On the other hand, if there are any nodes that have no links between any other developers, then they can be identified as someone not partaking appropriately in a teamwork environment. This can then be addressed by the team manager. Multiple projects are supported by the system by default (as a result of rigid database design), but it is challenging to crawl, split, and migrate all the necessary data manually.

However, the visualisation that is created is only best applied in larger projects where more nodes and links are visible. Smaller projects will not be able to benefit from the graph analysis since they will most likely be aware of any team issues already that are simply consolidated in the visualisation. An example of a smaller project is not always a team-based one. For instance, the Prolog Visualisation project by Saparamadu leveraged visualisations to help in "improved comprehension and analysis of logical programming"[13] to represent simple Prolog statements. Moreover, Petre and de Quincey make a point that scalability is paramount to software visualisations in general: "Selection and abstraction are crucial in addressing scale; the challenge is to find appropriate and meaningful selections and abstractions in order to provide a useful focused view"[2].

Lastly, it is important to consider the reasons for why a project may have inefficient workflow in their teams. For example, negative feedback from peers may result in developers in abandoning projects after only a few code reviews, leaving other teammates to pick up the pieces as Egelman et al. wrote: "In the process, developers may have negative interpersonal interactions with their peers, which can lead to frustration and stress; these negative interactions may ultimately result in developers abandoning projects"[3].

5 Use Cases

We believe that our system has several use cases and real-world applications and will make it easier to make data-driven decisions for a project. For instance, it is paramount in identifying inequitable workloads in a development team. Our visualisation helps project managers identify members of the team who are reviewing or being reviewed more than others. This is proven in Malik, "The statistical results showed that the agile practices of team autonomy and agile communication contributed to psychological empowerment that led to the innovative behaviour of agile teams"[9]. In addition to this, it helps enhance developer recognition and motivation and reduce resentment between members of a team. The size of each developer's contributions (i.e. the size of the nodes in the graph) are a useful quantitative measure of the work balance within the

team. John W. Koning Jr. says: "...lack of appropriate recognition and rewards can result ...or engineer resentment due to perceived unfair treatment"[6]. Furthermore, developers are encouraged to knowledge share more frequently as a result of information silos discovered through analysis of graph nodes. Node edge thicknesses help supplement how much intelligence is being shared between team members. Moreover, the code review process can be optimised more thoroughly and the necessary improvements in team dynamics can be made. Gelbard and Carmerli claim that: "Having higher levels of team dynamics was strongly associated with both forms of performance when the organization provided significant support to the project"[5]. Outside of team benefits, the visualisation also assists individual developers in evaluating their performance in a project and assessing whether they are putting in the adequate amount of work. Naturally, we believe developers will be more encouraged to work harder if they discover that their performance metrics are inferior to their colleagues. This is exemplified in Rasch and Tosi's study: "High achievement needs were directly related to both effort and perceived performance, whereas self-esteem and locus of control have a direct relationship to perceived performance"[12].

6 Conclusion

Visualising developer interactions in code reviews is a vital part of the development process to ensure efficient and healthy progress for a project. Through easy graph analysis, one can discover team dynamics and relationships that showcase the weaknesses in a team's workflows that can then be addressed by appropriate parties in a short timeframe.

In this paper, the several components that make up the efficient web system that was built to power and support this visualisation will be broken down. Several technologies will be evaluated and explored that made the development of this project a lot faster than it otherwise would have been. The purpose and related work of the system will also be discussed.

As future work, the plan is to make the system more automated and reduce the manual labour for the end-user during the crawling, splitting, and migration stages of the data collection phase. Graph propagation is another feature in plans to allow individual users to be the root of the graph and propagate to their contributor counterparts. Extension features also include node colour coding and project mixing and matching (i.e. multiple projects in a singular visualisation).

References

- [1] Alberto Bacchelli and Christian Bird. 2013. Expectations, outcomes, and challenges of modern code review. In *Proceedings of the ACM/IEEE 42nd International Conference on Software Engineering*.
- [2] Peter Eades and Kang Zhang. 1996. *Software visualisation*. Vol. 7. World Scientific.
- [3] Carolyn D. Egelman et al. 2020. Predicting Developers' Negative Feelings about Code Review. In *2020 IEEE/ACM 42nd International Conference on Software Engineering*. 174–185.
- [4] Daniel Russo et al. 2021. The Daily Life of Software Engineers during the COVID-19 Pandemic. In *43rd International Conference on Software Engineering: Software Engineering in Practice*. IEEE, 364.
- [5] Roy Gelbard and Abraham Carmeli. 2009. The interactive effect of team dynamics and organizational support on ICT project success. *International Journal of Project Management* 27, 5 (2009), 464–470.
- [6] John W Koning Jr. 1993. Three other R's: Recognition, reward and resentment. *Research-Technology Management* 36, 4 (1993), 19–29.

- [7] Phyllipe Lima, Jorge Melegati, Everaldo Gomes, Nathalya Steffany Pereira, Eduardo Guerra, and Paulo Meirelles. 2023. CADV: A software visualization approach for code annotations distribution. *Information and Software Technology* 154 (2023), 107089.
- [8] Abdul Majeed and Ibtsisam Rauf. 2020. Graph Theory: A Comprehensive Survey about Graph Theory Applications in Computer Science and Social Networks. In *School of Information and Electronics Engineering*. Korea Aerospace University.
- [9] Mohsin Malik, Shagufta Sarwar, and Stuart Orr. 2021. Agile practices and performance: Examining the role of psychological empowerment. *International Journal of Project Management* 39, 1 (2021), 10–20.
- [10] Shane McIntosh, Yasutaka Kamei, Bram Adams, and Ahmed E. Hassan. 2014. The Impact of Code Review Coverage and Code Review Participation on Software Quality: A Case Study of the Qt, VTK, and ITK Projects. In *Proc. of the Working Conference on Mining Software Repositories (MSR)*.
- [11] Rodrigo Morales, Shane McIntosh, and Foutse Khomh. 2015. Do Code Review Practices Impact Design Quality? A Case Study of the Qt, VTK, and ITK Projects. In *Proc. of the International Conference on Software Analysis, Evolution, and Reengineering (SANER)*.
- [12] Ronald H Rasch and Henry L Tosi. 1992. Factors affecting software developers' performance: An integrated approach. *MIS quarterly* (1992), 395–413.
- [13] Rosanne Saparamadu. 2023. Prolog Visualisation. In *Wellington Faculty of Engineering Symposium*.
- [14] Patanamon Thongtanunam, Chakkrit Tantithamthavorn, Raula Gaikovina Kula, Norihiro Yoshida, Hajimu Iida, and Ken-ichi Matsumoto. 2015. Who should review my code? A file location-based code-reviewer recommendation approach for Modern Code Review. In *IEEE 22nd International Conference on Software Analysis, Evolution, and Reengineering (SANER)*.
- [15] Yue Yu, Huaimin Wang, Gang Yin, and Tao Wang. 2016. Reviewer recommendation for pull-requests in GitHub: What can we learn from code review and bug assignment? *Information and Software Technology* (2016).